Degree budgets for binary summation systems: inverse families and arbitrary coordinate spaces

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Abstract

We study ordinary iterated last fall degree for explicitly presented binary two-summand elliptic-curve coordinate systems with Boolean field equations. For the inverse family over $\mathbb{F}_{2^n}$ we give a written proof of the exact value 3 for odd n, and $\max(3, n/2+1)$ for even n. The argument translates classical cyclic nonadjacent representations into degree-charged polynomial rewrites and a full-closure lower obstruction. For arbitrary coordinate spaces of dimensions k, ℓ and any target, a fiber-selector argument gives LFD $\leq \min(k, \ell) + 2$. Together these give linear worst-case growth under a dimension-at-most budget, with explicit lower and upper constants. A seeded campaign outside the scaled-subfield class gives 613 distinct systems with exact finite degree computations; its records are not included in the public repository. We distinguish written proofs, kernel-checked components, and finite arithmetic throughout. An internal referee report on the proofs exists; independent external proof and novelty review remain outstanding. No useful-relation cost improvement or generic ECDLP complexity claim is made.

1 Scope and conventions

Let $K = \mathbb{F}_{2^n}$ and

$$E: \quad y^2 + xy = x^3 + ax^2 + b, \qquad b \neq 0.$$

For a finite target with x-coordinate t, the third summation polynomial is

$$S_3(x, y, t) = (xy + xt + yt)^2 + xyt + b. \tag{1}$$

Here x, y denote summand x-coordinates, not the two coordinates of one point. Write $x = \sum \beta_i X_i$, $y = \sum \gamma_j Y_j$ for explicit $\mathbb{F}_2$-bases of spaces W_1, W_2 of dimensions k, ℓ. Expand (1) in an ambient field basis and reduce each coordinate modulo $Z^2 - Z$ for every input bit. These reduced coordinate polynomials, together with the original Boolean equations, form our *normalized* input F. They have bidegree at most $(1, 1)$. The ordinary unreduced coordinate polynomials give a different, raw presentation. Normalization is part of the declared presentation; it is not a free step inside an ordinary closure.

For a finite input $F \subset A = \mathbb{F}_2[Z_1, \ldots, Z_r]$, let $V_{F,D}$ be the least coefficient-linear subspace containing inputs of degree at most D, and closed under $h \mapsto gh$ whenever $\deg g + \deg h \leq D$. This multiplication applies to newly derived rows. We use the ordinary total degree of polynomial representatives. Define

$$\mathrm{LFD}(F) = \min\{d : \ h \in \langle F \rangle \Longrightarrow h \in V_{F,\max(d,\deg h)}\}.$$

The convention permits zero and is meaningful for proper ideals. It is not a degree at which linear solving happens to stop. These closure spaces and the largest-strict-fall interpretation are established prior art [1, 2].

Remark 1 (Presentation dependence). Degree-preserving invertible affine changes preserve these closures after transport of the inputs. Scalar extension commutes with every finite-degree closure: row spans and all multiplication conditions are defined by linear equations, and field extension is exact. Input row operations preserve LFD when the spans of the inputs of degree at most D agree *for every* D. Equality of ideals or of unfiltered row spans alone is insufficient. The specific transformations below satisfy this stronger condition.

Our theorems concern polynomial coordinate relations. Rational lifts and independent relations for discrete logarithms impose additional conditions. We do not identify homogeneous first fall degree with LFD or a collection cost. Prime-field and larger-chain artifacts remain available in the repository but are not hypotheses of the results here.

2 A bound for arbitrary coordinate spaces

Theorem 2. *For every K, a, b as above, every $t \in K$, and arbitrary spaces W_1, W_2, the normalized presentation satisfies*

$$\mathrm{LFD}(F) \le \min(k, \ell) + 2.$$

There is no subfield, Frobenius-invariance, genericity, or target-order assumption. Rationality of the target is unnecessary for this algebraic statement.

Proof. Assume $k \le \ell$ and put $D = k + 2$. Fix $x \in W_1$. As a polynomial in y, (1) becomes

$$(x + t)^2 y^2 + xt\, y + (xt)^2 + b. \tag{2}$$

This is affine-linear on Boolean coordinates of W_2. It is a nonzero scalar polynomial: simultaneous vanishing of its quadratic and linear coefficients forces $x = t = 0$, leaving the nonzero constant b. Thus a coordinate fiber is empty, a point, or an affine $\mathbb{F}_2$-line, since there are at most two scalar roots.

For $\alpha \in \mathbb{F}_2^k$ let $\chi_\alpha = \prod_{\alpha_i=1} X_i \prod_{\alpha_i=0}(1 - X_i)$. Its degree is k. The Boolean inputs derive $\chi_\alpha(X_i - \alpha_i)$ at budget $k + 1$ by explicitly factoring out $X_i^2 - X_i$. Multiplying a normalized coordinate input by χ_α costs at most $k + 2$. Substitution $X = \alpha$ in this product changes it by a sum of those selector defects times affine-linear polynomials in Y, also costing at most $k + 2$. All affine fiber rows multiplied by the selector are therefore in $V_{F,D}$.

Gaussian elimination of this affine system uses coefficient combinations only. For an empty fiber it derives χ_α. For a point it derives $\chi_\alpha(Y_j - c_j)$. For a line choose a coordinate T differing at its two points; it derives

$$\chi_\alpha(Y_j - c_j - d_j T), \qquad c_j, d_j \in \mathbb{F}_2. \tag{3}$$

These rows have degree at most $k + 1$ and can be multiplied by T at budget D. The input $T^2 - T$ times χ_α also costs D.

Let S be the complete Boolean zero set and $q : A \to \mathbb{F}_2^S$ evaluation. The point-fiber functions χ_α, and the line-fiber functions $\chi_\alpha, \chi_\alpha T$, form a basis of functions on S. Use these degree-at-most-$k + 1$ representatives to define a linear section s of q. For each variable Z and each such basis function h, the row

$$Zs(h) - s(q(Zs(h)))$$

is in $V_{F,D}$: for X_i use the selector defect, possibly times T; for Y_j use (3), its T multiple and the Boolean correction. The constant correction $1 - s(q(1))$ is the sum of the empty-fiber selectors, because $\sum_\alpha \chi_\alpha = 1$ exactly.

For completeness, these corrections imply all-degree reduction, not just a stopping rule. Induct on monomial degree r. The previous correction has degree at most $\max(r-1, k+1)$; multiplying it by the next variable costs at most $\max(r, D)$. Add the available section correction. Hence

$$p - s(q(p)) \in V_{F,\max(D,\deg p)}$$

for every polynomial p, by linearity and monotonicity of budgets. If $p \in \langle F \rangle$ then $q(p) = 0$, so this is the required LFD property. This also covers $S = \varnothing$, when $s = 0$. Swapping the coordinate blocks gives the minimum in the statement. □

The raw presentation can derive every Boolean-normalized coordinate row at budget four by the Boolean correction identities. Its bound is consequently $\max(4, \min(k, \ell)+2)$, with this preprocessing budget charged. We do not assert equality between raw and normalized LFDs.

3 The inverse presentation

Take $b = 1$, $R = (0, 1)$, and initially $W_1 = W_2 = K$. The normalized inputs are the coordinates of $xy + 1$ and the Boolean field equations.

Lemma 3 (Moore coordinates)**.** *After extension of coefficient scalars to K and an invertible homogeneous linear change of variables, the filtered input spans equal those of*

$$u_i^2 - u_{i+1}, \quad v_i^2 - v_{i+1}, \quad u_i v_i - 1, \qquad i \in \mathbb{Z}/n\mathbb{Z}. \tag{4}$$

Thus these operations preserve ordinary iterated LFD.

Proof. For a basis β_j put $u_i = \sum_j \beta_j^{2^i} X_j$, and similarly define v_i. The Moore matrices are invertible because their columns are $\mathbb{F}_2$-linearly independent. The square-chain equations are invertible combinations of the Boolean equations. Applying the Moore matrix to the output coordinates of $xy + 1$ gives $u_i v_i + 1$. This also describes the output of the raw $S_3 = (xy + 1)^2$ after the declared Boolean normalization, by an invertible output Frobenius.

In these coordinates the $3n$ quadratic tops are distinct monomials: u_i^2, v_i^2 and $u_i v_i$. They are independent. Every nonzero row in their span has degree two. Thus the filtered input spans agree even below degree two, as required in Remark 1. □

Modulo the two square chains, each monomial has a unique squarefree normal form in each block, obtained by carrying squares cyclically. Degree never increases; a positive exponent never becomes zero. The remaining equations restrict the zero set to the inverse graph $u_i = x^{2^i}, v_i = x^{-2^i}$ with $x \in K^*$. The chain quotient is the reduced algebra of functions on K^2, so a polynomial is in the full ideal of (4) exactly when it vanishes on this graph. Put $M = 2^n - 1$. A squarefree monomial with positive and negative bit words e, f evaluates as the character $x^{e-f \bmod M}$. The M characters are linearly independent over K: a polynomial of degree below M vanishing on K^* is zero. Ideal membership is therefore zero coefficient sum in each residue class.

4 Degree-charged signed-digit normalization

We give the combinatorial argument to specify the polynomial budget. The cyclic nonadjacent uniqueness theorem itself is classical, due to Preparata [4]; we make no originality claim for it.

Lemma 4. *For $n \geq 2$, every monomial in (4) reduces to a cyclic nonadjacent signed word, without increasing weight, at budget* $\max(3, \deg p)$.

Proof. Square-chain reduction first makes the two blocks squarefree. Cancel opposite signs at one position using $u_i v_i = 1$. The following additional differences are derivable at degree three, with indices modulo n:

$$u_{i+1} v_i \leftrightarrow u_i, \qquad u_i v_{i+1} \leftrightarrow v_i, \qquad u_i u_{i+1} \leftrightarrow v_i u_{i+2},$$

and the last rule with signs reversed. For example,

$$u_i(u_i v_i - 1) - v_i(u_i^2 - u_{i+1}) = u_{i+1} v_i - u_i,$$
$$v_i(u_{i+1}^2 - u_{i+2}) - u_{i+1}(u_{i+1} v_i - u_i) = u_i u_{i+1} - v_i u_{i+2}.$$

Each displayed product has degree at most three before reduction. A quadratic derived row in a monomial context of total weight w can then be multiplied by the remaining $w - 2$ factors at budget w.

Mixed adjacent signs reduce weight. Equal adjacent signs use $(+, +, 0) \mapsto (-, 0, +)$ or its sign reverse. If the destination is occupied, cancellation or a square-chain carry reduces weight. If the position just left of the pair was occupied, it becomes an opposite-sign adjacency, again reducing weight. Otherwise the pair is left-isolated. The rewrite fixes an opposite-sign digit behind the moving digit. If the next position is empty, the number of occupied adjacencies decreases; if it has opposite sign, weight decreases; if it has equal sign, propagate the same rewrite. The moving pair advances by two positions. It must stop within n steps: revisiting a starting position would already encounter one of the fixed opposite-sign digits, causing cancellation or a mixed adjacency. Thus every outer phase decreases lexicographically the weight and the number of occupied adjacencies. Both are nonnegative. Normalization terminates without exceeding the original weight.

For $n = 2$, an equal adjacent pair rewrites to $v_i u_i$ (or its sign reverse), then cancels; mixed adjacency reduces immediately. These handle the short-cycle coincidences in the rule indices. □

Lemma 5 (Classical cyclic rigidity)**.** *For lengths $n \geq 2$, cyclic nonadjacent signed words of odd length have distinct residues modulo M. For $n = 2d$, the only distinct words with the same residue are the two sign/rotation versions of*

$$(+, 0, +, 0, \ldots, +, 0), \qquad (0, -, 0, -, \ldots, 0, -).$$

They have weight d.

Proof. For $n \geq 2$ compare words $a_i, b_i \in \{-1, 0, 1\}$ with the same residue. Their absolute values are below M. Their difference is Mc_0 with $c_0 \in \{-1, 0, 1\}$. Define carries

$$2c_{i+1} = c_i + a_i - b_i.$$

They are integers: the congruence $Mc_0 = \sum_i 2^i (a_i - b_i)$ implies that $c_0 + \sum_{j<i} 2^j (a_j - b_j)$ is divisible by 2^i. It also gives $c_n = c_0$. The recurrence keeps them in $\{-1, 0, 1\}$. An exit from carry zero requires opposite nonzero digits; both next digits then vanish, contradicting integrality of the next carry. By periodicity, there is no entry into zero either. Carries cannot switch between 1 and -1, which would require digit difference of magnitude three. They are constant. Carry zero gives identical words; carry 1 requires alternating $(a_i, b_i) = (1, 0), (0, -1)$ by nonadjacency. Carry -1 reverses signs. The alternating pattern requires even length. The $n = 1$ case is handled directly in Theorem 9, rather than by this lemma. □

Lemma 6. *For $n = 2d$, the two exceptional monomials $U = \prod_{i \text{ even}} u_i$, $V = \prod_{i \text{ odd}} v_i$ are connected at budget* $\max(3, d + 1)$.

Proof. Replace u_0 by u_1v_0 using the carry row backward, increasing weight to $d+1$. Apply $u_1u_2 \mapsto v_1u_3$, then $u_3u_4 \mapsto v_3u_5$, continuing through $u_{n-3}u_{n-2} \mapsto v_{n-3}u_{n-1}$. Finally $u_{n-1}v_0 \mapsto v_{n-1}$. Weight never exceeds $d+1$. The short case $d=1$ is the carry relation itself. Rotation and sign exchange give the other pair. □

5 The full-closure lower obstruction

Lemma 7 (Positive-word isolation)**.** *If T is a cyclic nonadjacent set of positions and $|T| < n/2$, its positive word is the unique signed representation of its residue having weight at most $|T|$.*

Proof. For $T = \varnothing$ the weight condition allows only the empty word. Otherwise cancel any simultaneous positive and negative digit, reducing weight. Compare digits $t_i \in \{0,1\}$ of T with arbitrary signed digits s_i. Their cyclic carry recurrence is $2c_{i+1} = c_i + t_i - s_i$. The initial carry is zero or one because their weighted difference is a multiple of M strictly between $-M$ and $2M$. Integrality keeps all carries in $\{0,1\}$.

A proper run of carry one starts at $t_a = 1, s_a = -1$ and ends at $t_b = 0, s_b = 1$. Between them $s_i = t_i - 1$. In the inclusive interval $[a,b]$, let p be the number of positive digits of T and z the number of zeros. Nonadjacency and the endpoints give $z \geq p$. The new signed word has weight $z+1 > p$ on this interval. Outside carry-one runs the digits agree. Carry one on the entire cycle instead gives the negative complement of weight $n - |T| > |T|$. Any different representation therefore has larger weight. □

Proposition 8. *For even $n = 2d \geq 6$, the system (4) has LFD at least $d+1$.*

Proof. The row $U - V$ from Lemma 6 belongs to the ideal, has degree d, and has coefficient one at U. We show that every normal image of $V_{F,d}$ has coefficient zero at U.

For any ideal row h of normal degree at most $d-1$, the monomial U/u_i for even i has a positive nonadjacent word of weight $d-1$. Lemma 7 isolates its character among all monomials in h. Vanishing on the inverse graph forces its coefficient to be zero.

Multiplication of h by a v variable cannot produce the pure-u monomial U: square-chain normalization of a positive v exponent never makes it zero. For multiplication by u_i at even i, the only squarefree preimage of weight at most $d-1$ is U/u_i. A carry would erase the required bit at i unless it wrapped around the full cycle, which requires too many bits. At odd i, a preimage must have the bit at i and all other even bits except $i+1$, hence weight at least d; a longer carry erases another required even bit. Thus no allowed variable product creates a coefficient at U.

Every input has degree at most two and zero U coefficient, since $d \geq 3$. Addition and scalar multiplication preserve the condition. Variable-product stability implies stability under every allowed multiplier by factoring its monomials into variables, with all intermediate budgets bounded by the final budget. Square-chain normal form never increases degree, and all derived rows remain ideal rows. The least-space property of the full iterated closure now gives the claim. In particular, $U - V \notin V_{F,d}$, which excludes every LFD bound at most d. □

Theorem 9. *For $K = \mathbb{F}_{2^n}$, $E : y^2 + xy = x^3 + ax^2 + 1$, $R = (0,1)$ and the full-field coordinate base, the normalized presentation has*

$$\mathrm{LFD}(F) = \begin{cases} 3 & n \text{ odd}, \\ \max(3, n/2+1) & n \text{ even}. \end{cases}$$

Proof. For $n \geq 2$, reduce every monomial of any ideal polynomial by Lemma 4. Odd uniqueness, or the even bridge at $L = \max(3, n/2+1)$, leaves one chosen representative per residue.

Character independence makes the remaining coefficient sums zero. Every step costs at most $\max(L, \deg p)$, proving the full-ideal upper bound in all degrees. Proposition 8 gives the even lower bound when $n \geq 6$.

For all $n > 1$, degree-two input tops are independent, so their span has no nonzero row below degree two. It is already closed at budget two. The ideal row $u_{i+1}v_i - u_i$ is absent from this span, since its off-diagonal quadratic top is absent. It is derived at degree three. Hence LFD is at least three, also covering even $n = 2, 4$.

For $n = 1$, the same carry derivation gives $uv - u$, which is absent from the degree-two span: the forced coefficient of the $uv - 1$ input leaves an incorrect constant. Together with $uv - 1$, it yields $u - 1$ at degree three, and similarly $v - 1$. These generate the maximal ideal of $(1, 1)$; substitution reduction then gives its full-ideal LFD upper bound three. Finally apply Lemma 3. □

Under the independent coordinate-input, square-zero truncated Koszul/Frobenius convention, the first fall degree is three. Indeed the independent quadratic tops are u_iv_i modulo all variable squares. There is no degree-two relation, while $u_i(u_iv_i) = 0$ is a degree-three syzygy and the trivial Koszul/Frobenius rows start in weighted degree four. This statement uses the declared independent list. Retaining redundant ambient output rows can instead produce FFD two; LFD still agrees because the filtered input spans agree. Our data record both conventions for scaled-subfield cases.

6 Proper bases and worst-case order

Corollary 10. *Let $L = \mathbb{F}_{2^d} \subset K$, $\alpha \neq 0$, $s \in L^*$, and $c = \alpha^2 s$. Use common space $W = \alpha L$, curve coefficient $b = c^2$, and target $R = (0, c)$. The normalized LFD is the formula in Theorem 9 with d in place of n.*

Proof. The substitution $x = \alpha u, y = \alpha v$ changes the target-zero scalar equation into the coordinates of $uv + s$, up to an invertible output Frobenius and scalar. Scale one coordinate block by s to obtain $uv + 1$ over L. An ambient output basis adapted to $\alpha^4 L$ leaves d independent coordinates. Other ambient bases give the same filtered span: every nonzero combination of these outputs has a nonzero quadratic top, as multiplication $L \times L \to L$ is onto. Boolean rows add independent square tops. Apply the preceding transport argument. □

The space αL is Frobenius invariant exactly when $\alpha \in L$. This includes proper subfields and noninvariant dilations. Any fixed $b \neq 0$ allows a compatible choice $\alpha = b^{1/4}, s = 1$; the target remains the order-two point. When $\alpha = s = 1$, $a \in L$, and $[K : L]$ is even, every nonzero $x \in L$ lifts rationally. Dividing the curve equation by x^2 gives an Artin–Schreier equation with right side in L, whose absolute trace to $\mathbb{F}_2$ vanishes in K. Direct addition gives $x(R - P) = 1/x$. This proves rational decomposability for this specialization, not independence or usefulness of relations in a large prime-order group.

Corollary 11 (Dimension-budget envelope)**.** *Let $M_{\leq}(k)$ be the supremum of normalized two-summand LFD over all finite binary fields, nonsingular curves of the stated form, finite rational targets, and two arbitrary coordinate spaces of dimensions at most k. The same notation with a common-space restriction has the same following bounds. For every integer $k \geq 6$,*

$$\lfloor k/2 \rfloor + 1 \;\leq\; M_{\leq}(k) \;\leq\; k + 2.$$

Thus its worst-case growth is $\Theta(k)$.

Proof. Theorem 2 bounds every member of either class by $k + 2$. Set $d = 2\lfloor k/2 \rfloor$, choose $K = \mathbb{F}_{2^{2d}}$, common proper subfield space $L = \mathbb{F}_{2^d}$, and $a = 0, b = 1, R = (0, 1)$. Corollary 10 gives the lower value $d/2 + 1$. □

This is order sharpness over a class allowing arbitrary spaces. Its witness is structured. It neither gives a common degree for individual spaces nor the exact worst-case constant. For odd k the witness has dimension $k-1$; the at-most quantifier is essential. No monotonicity claim about the exact-dimension maximum is used. No arbitrary-$m \geq 3$ consequence is asserted.

7 Finite evidence outside the structured family

The records of this section are not included in the public repository `isogeny-labs/lfd-summation-systems`; the numbers below are reported from the original campaign and cannot be checked from that repository. The new deterministic campaign uses seed 20261030. For $k = 2, \ldots, 8$ take ambient degrees in $\{k+1, k+3, 2k\}$ and four canonical random spaces per degree, rejecting every scaled subfield. Such rejection is exact: if $k \mid n$, a dimension-k space is a scaled subfield precisely when all basis ratios are fixed by 2^k Frobenius. Use $(a, b) = (0, 1), (1, 1), (z, z+1)$, with z the generator of the polynomial basis of K (integer encodings 2 and 3), and three target categories. The resulting 702 geometric cases consist of 240 order-two, 240 nonzero random rational, and 222 planted rational-decomposition targets. Eighteen unavailable planted cases are recorded as skips. All targets and planted sums are checked by exact group arithmetic.

Systems are deduplicated by their literal normalized input rows and variable count, after dropping zero and duplicate rows. We do not replace row spans by a Gaussian basis that might make low-degree consequences free inputs. There are 613 distinct systems and no remaining resource caps:

LFD	0	1	2	3	4
Distinct systems	7	1	50	381	174

At every scanned degree the exact arithmetic verifies a containing space closed under permitted products. A local evaluation-rank certificate bounds the full ideal; strict falls then determine the exact LFD. Enumerating fixed-coordinate affine fibers provides the solution set, checked separately against scalar S_3. The replay regenerates all geometry and recomputes every saved degree profile. These matrix computations are Python evidence, not concrete Lean certificate imports.

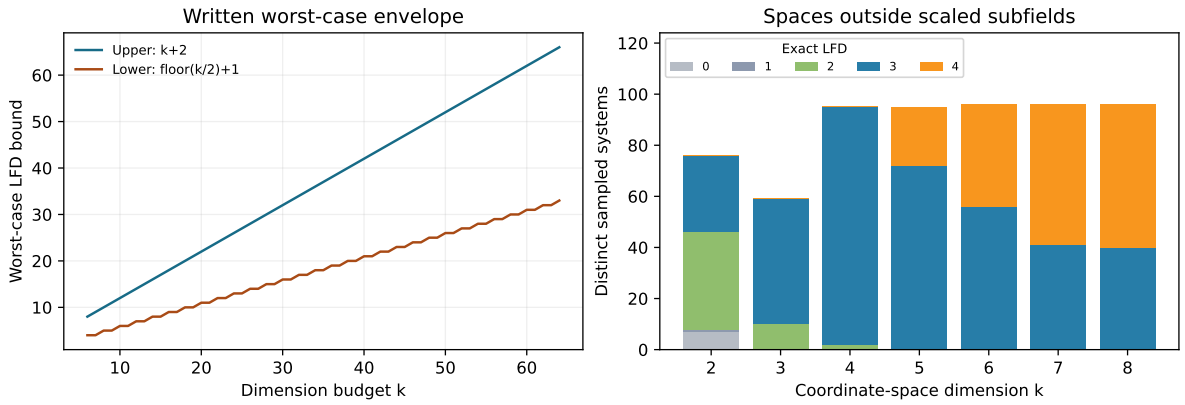


Figure 1: Left: written bounds on the worst-case dimension-budget quantity, not exact values. Right: exact LFD histogram by dimension for the new seeded systems outside the scaled-subfield class. No population inference is made.

The data store algebraic tuples, rational coordinate tuples and unordered multisets separately. For example, 167 order-two cases have polynomial solutions but only 141 have rational

coordinate decompositions; all 222 planted cases pass their rational existence checks. Independent relation rank remains unknown. The samples cannot prove a constant-four universal bound, which the structured growing family already excludes.

The separate existing artifact contains 1,803 geometric cases / 1,474 distinct systems with $m = 2, \ldots, 9$, all now exact after resolving 29 earlier caps. It also contains 587 scaled-subfield/dilation cases and inverse-model traces through $n = 12$. These campaigns have overlapping geometry and different presentations; their counts are not added as independent samples. The original prime-field model experiments did not establish a net decrease in measured useful-relation collection cost.

8 Related work and verification boundary

Huang, Kosters, Yang and Yeo established the LFD and Weil-descent framework used here [1]. Caminata and Gorla compare LFD with solving degree and homogeneous invariants [2]. Moore coordinates and cyclic Frobenius chains are standard descent machinery, not new constructions. Gorla, Mueller and Petit give improved LFD bounds for univariate fake Weil descent and HFE systems [3], especially Section 6, Theorem 11 and Corollary 1. Their input is a univariate system with field equations; a direct application to this two-variable inverse presentation requires a degree-budget-preserving reduction, not just a parameterization by inversion. Their result is not an exact calculation for the presentation here. This distinction does not establish priority for our consequence.

Preparata's cyclic nonadjacent representation theorem already supplies the odd uniqueness and even exceptional patterns [4]. Standard signed-digit minimality is also discussed by Bosma [5]. The candidate contribution is the explicit ordinary iterated-closure budget, full lower obstruction, and its curve/base specialization. General FFD/LFD separation, ECDLP descent applications, subfield factor bases, and CAS proof certification are established research topics. The repository's source-level novelty audit records additional overlapping work and which primary texts were read. An exact novelty claim requires independent specialist review.

The pinned Lean 4.19.0/mathlib artifact audits 513 declarations (424 when this draft was first written) without project axioms, proof placeholders or native-decide overrides. Among the checked components are the ordinary closure definitions, degree-safe affine transport, local-to-global reduction theorem, scalar binary fiber bounds, bounded cyclic carries constructed from modular equality, their rigidity, local degree-three identities, and the all-length even product bridge. Additional checked links prove arbitrary-dimensional coordinate and Frobenius identities, invertibility of the actual Moore matrix, both directions of finite coefficient-field extension for full closures and exact LFD, and exact affine/input transport with explicit budgets. The main normalization, coefficient lower obstruction, final Moore input-filtration specialization and arbitrary-space selector specialization remain written proofs. Counting checked supporting declarations does not make Theorems 2 and 9 fully kernel-checked. Separate kernel results include all-k prime two-summand FFD classification and lower bounds for all-length one-bit split systems; neither supplies universal larger-m arbitrary-base sharpness.

9 Reproduction and unresolved questions

The original source package included this manuscript, figures, exact data, source/data SHA256 manifests, pinned formal sources and audit logs. Of its scripts, only `verify_lean_fall_degrees.py` is in the public repository, together with the formal sources; the commands below, other than that one, refer to the original package and do not run from the public repository. From the original package:

```
python scripts/reproduce_ic_paper.py --lake /path/to/lake
```

This command runs the full proof-component/data/paper pipeline in an isolated source copy, including fresh baseline and frontier arithmetic. The report lists the main theorem's open formal obligations. Adding `-require-complete-proof` makes the command exit with failure status two while these obligations remain open. It does not obtain independent review. Individual components can also be run directly:

```
python scripts/build_ic_paper.py
python scripts/run_ic_arbitrary_bases.py --check-only
python scripts/verify_ic_positive_naf.py --check-only
python scripts/verify_ic_inverse_parity.py
python scripts/run_ic_inverse_subfields.py
python scripts/verify_ic_binary_m2_uniform.py
python scripts/verify_ic_research_artifact.py --replay-inverse-traces
python scripts/verify_lean_fall_degrees.py --lake /path/to/lake
```

The last command requires the pinned Lean toolchain and mathlib dependencies; they and copyrighted literature downloads are excluded from the source zip. Building the PDF requires pdfLaTeX and matplotlib. The new positive-word check exhausts signed words through length 12 and compares all minimum-weight representations; it supports Lemma 7, rather than proving its universal statement computationally.

Outstanding mathematical questions include the exact worst-case constants, sharp values for unstructured individual spaces and arbitrary targets, and larger-m systems with unrestricted finite partial sums. Independent proof review, publication novelty, authorship and venue metadata are not settled by these experiments or formal components. This manuscript is an internal review package, not a claim of submission readiness.

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